Sungyeon Kook

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Education

Korea University — Seoul, Korea
M.S. (Master of Science) in Electrical Engineering — Mar. 2024 – Feb. 2026
B.S. (Bachelor of Science) in Electrical Engineering — Mar. 2020 – Feb. 2024

Research Experience

Korea Institute of Science and Technology Information (KISTI) — Daejeon, Korea
Post-Master Researcher, Quantum Network Research Center (Advisor: Dr. Ilkwon Sohn, Dr. Wooyeong Song) — Mar. 2026 – Present

Korea University — Seoul, Korea
Graduate Research Assistant, Communications and Information Systems Laboratory (Advisor: Prof. Jun Heo) — Mar. 2024 – Feb. 2026

Korea Research Institute of Standards and Science (KRISS) — Daejeon, Korea
Visiting Student Researcher, Center for Superconducting Quantum Computing System (Advisor: Dr. Hwan-Seop Yeo) — Aug. 2024 – Aug. 2025

Korea University — Seoul, Korea
Undergraduate Research Assistant, Communications and Information Systems Laboratory (Advisor: Prof. Jun Heo) — Jan. 2023 – Feb. 2024
Undergraduate Research Assistant, Quantum Control Laboratory (Advisor: Prof. Mahn-Soo Choi) — Jul. 2023 – Aug. 2023

Research Interests

Quantum Information, Quantum Error Correction, Fault-Tolerant Quantum Computing, Error Correcting Codes

Publications (* FIRST AUTHOR, † CORRESPONDING AUTHOR)

[1] Changyeol Lee*, **Sungyeon Kook***, Wooyeong Song, Kwangil Bae, Wonhyuk Lee, and Ilkwon Sohn†
"Efficient Quantum Modular Reduction: Crandall Reduction and Its Fault-Tolerant Resource Analysis,"
Submitted to *Scientific Reports* | arXiv:2608.11563.

[2] Wooyeong Song*, **Sungyeon Kook**, Wonhyuk Lee, and Ilkwon Sohn†
"Asymmetry-Aided Measurement-Based Quantum Repeaters and Distributed Quantum Computing with a Decoder-Free Client," Submitted to *npj Quantum Information* | arXiv:2607.11238.

[3] **Sungyeon Kook***, Yujin Kang, Ilkwon Sohn, and Jun Heo†
"Low Spatial Cost CCZ Magic State Factory,"
Under review at *Quantum Information Processing* | arXiv:2606.24170.

Conference Presentations

[1] Asymmetry-aided Measurement-Based Quantum Repeaters and Distributed Quantum Computing with a Decoder-Free Client
- **Asian Quantum Information Science Conference (AQIS 2026)** [url] — Co-author | Aug. 2026
- Optics and Photonics Congress (OPC 2026) — Co-author | Jul. 2026

[2] Low Spatial Overhead CCZ Magic State Factory
- 2026 Annual Meeting of the Quantum Information Society of Korea — Poster | Feb. 2026
- International Conference on ICT Convergence (ICTC 2025) [url] — Poster | Oct. 2025

[3] Resource-efficient Logical Qubit Architectures for Magic State Consumption
- **Quantum Error Correction Conference (QEC 2025)** [url] — Poster | Aug. 2025

[4] Quantum Circuit Scheduling of Checkerboard Architecture Based on Lattice Surgery
- 2025 KICS Summer Conference — Oral | Jun. 2025

[5] Greedy DAG Scheduling for Quantum Circuit Transpile
- Joint Conference on Communications and Information (JCCI 2025) — Oral | Apr. 2025

[6] Comparison and Analysis of Surface Code Mapping Methods on Heavy Hexagon Structure
- 2025 KICS Winter Conference — Oral | Feb. 2025

[7] Limited Connectivity Quantum Architecture: Enhancing Surface Code Implementation through Routing Qubits
- International Conference on ICT Convergence (ICTC 2024) [url] — Oral | Oct. 2024

[8] Magic State Distillation with Reduced Time Cost
- **IEEE International Conference on Quantum Computing and Engineering (QCE 2024)** [url] — Co-author | Sep. 2024

[9] Implementation of SC-17 Decoding Using MWPM Decoder
- 2024 KICS Summer Conference — Oral | Jun. 2024

Research Projects

Fault-Tolerant Resource Analysis for Quantum Modular Reduction **[Pub. 1]** *2026–Present*
Advisor – Dr. Ilkwon Sohn *KISTI*

- Derived circuit-specific logical-error budgets from qubit count and T-depth using the KQ formalism, and mapped them to required rotated surface-code distances under target circuit failure probabilities and physical error rates.
- Formulated an end-to-end fault-tolerant runtime model combining T-state distillation and decoder backlog, quantifying how T-depth reductions affect code distance, magic-state latency, decoding load, and execution time.

Fault-Tolerant Code Switching and Decoder-Free Quantum Networking **[Pub. 2] [Conf. 1, 9]** *2024–Present*
Advisor – Prof. Jun Heo, Dr. Wooyeong Song *Korea University, KISTI*

- Developed multi-round gauge fixing between asymmetric Bacon–Shor and rotated surface codes, preserving logical Pauli algebra while achieving distance-m switching for $\mathrm{BS}(m \times 3) \to \mathrm{SC}(m \times m)$ versus distance 1 for switching.
- Constructed a centralized decoding framework jointly processing Bacon–Shor transmission, surface-code computation, and return-transmission syndromes, enabling global QEC without real-time client decoding.

Low-Spatial-Cost CCZ Magic-State Factory **[Pub. 3] [Conf. 2, 3, 8]** *2024–Present*
Advisor – Prof. Jun Heo *Korea University*

- Converted the $[[8, 3, 2]]$ CCZ distillation circuit into eight Z-basis $\pi/8$ Pauli-product joint measurements, eliminating eight $|0\rangle$ auxiliary qubits and reducing persistent logical qubits from 12 to 4.
- Established single-fault detection for the joint-measurement construction and derived logical-error models for the two-level factory, accounting for surface-code logical faults and injected-state errors.
- Constructed a compact surface-code implementation using four-qubit twisted ZY joint measurements, designing a two-level CCZ factory with 71.4% lower spatial cost and 37.7% lower space-time volume under pipelined scheduling.

Architecture-Aware Quantum Circuit Mapping and Scheduling **[Conf. 4, 5, 6]** *2024–2025*
Advisor – Prof. Jun Heo *Korea University*

- Analyzed flag-qubit–SWAP trade-offs for surface-code mapping on heavy-hexagon architectures and formulated a QODG-based scheduler for checkerboard lattice-surgery architectures.
- Developed a Greedy DAG scheduling pass in Qiskit using topological dependencies and bitmask-based qubit-conflict tracking to construct parallel gate layers and expose additional gate-level parallelism.

Crosstalk-Aware Surface-Code Scheduling via Graph Coloring *2024–2025*
Advisor – Prof. Jun Heo *National R&D Project | Korea University*

- Modeled CNOT incompatibilities induced by superconducting crosstalk as a graph-coloring problem, assigning operations to parallel groups that prevent nearby conflicts while maximizing stabilizer-measurement parallelism.
- Applied the scheduler to a distance-24 surface-code circuit with IBM-inspired hardware parameters, organizing the stabilizer CNOT network into circuit depth 8 while satisfying the imposed crosstalk constraints.

Experimental Quantum Error Correction on Superconducting Qubits **[Conf. 7]** *2024–2025*
Advisor – Dr. Hwan-Seop Yeo *National R&D Project | KRISS*

- Characterized superconducting qubits through spectroscopy, T_1/T_2 measurements, DRAG calibration, and randomized benchmarking, extracting hardware-level parameters for quantum error-correction experiments.
- Developed connectivity-aware CNOT/SWAP operations and experimentally implemented stabilizer-projection circuits on superconducting hardware, validating the required quantum operations through quantum state tomography.

Variational Quantum Algorithms *2023*
Advisor – Prof. Mahn-Soo Choi *Korea University*

- Implemented VQE and QAOA circuits with classical parameter optimization and developed quantum-kernel models using Qiskit and PennyLane for variational quantum optimization and machine-learning tasks.

Professional & Teaching Activities

IBM Qiskit Advocate — Aug. 2025 – Present
Teaching Assistant, Data Science and Artificial Intelligence — Fall 2024
Co-Founder and Treasurer of QUantum Information Club at Korea University (QUICK) — Jan. 2023 – Dec. 2023
Organizer of Qiskit Fall Festival 2023 — Oct. 2023

Honors & Awards

Outstanding Academic Achievement, Korea University — Spring 2023
Excellence Award (Korea University President's Award), Creative Challenger Program, Korea University — 2022
Encouragement Award, Startup Express Summer Session, Korea University — 2022

Technical Skills

Programming: Python, C, Matlab, **Quantum Software:** Qiskit, Stim, PyMatching, PennyLane